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Urinary stones resembling uric acid stones

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A 52-year-old man was referred for the evaluation of mild renal failure with severe hypertension (220/130 mmHg), which was resistant to combination therapy of a calcium channel blocker and an angiotensin II receptor blocker. Although he reported no past history indicating urolithiasis, urinalysis revealed plenty of round brownish stones (Figure 1A). The stones resembled uric acid (UA) stones (Figure 1B), but most of them had spicules radiating from the centre and rough surface, which are not prominent in UA stones. Furthermore, the stones were friable compared with UA stones. From these characteristics, the stones were recognized as 2,8-dihydroxyadenine (2,8-DHA) stones [1,2]. The diagnosis of type II adenine phosphoribosyltransferase (APRT) deficiency was confirmed by the genomic analysis that revealed APRT*J homozygote (Figure 2) [3]. The treatment by allopurinol with a low purine and low sodium diet diminished the stones and reduced the systemic blood pressure to 140/80 mmHg. In APRT deficiency, the oxidation to 2,8-DHA is the met-

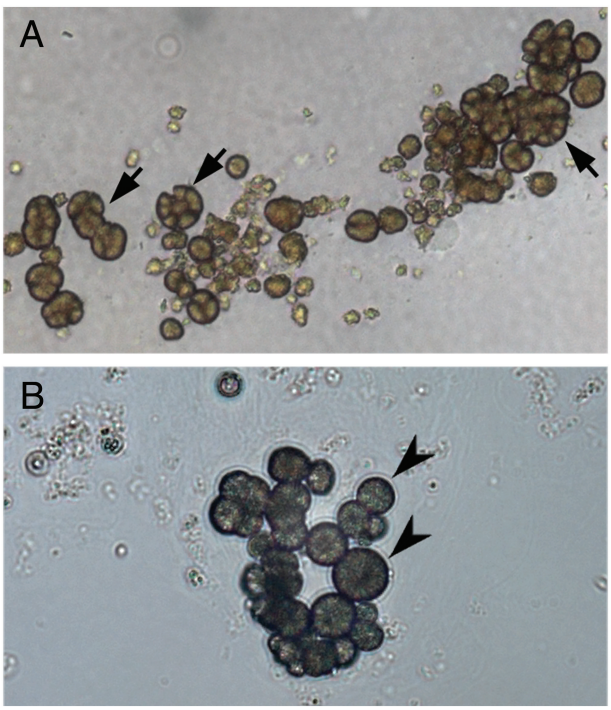


Fig. 1. Light microscopy of urine sediment revealing round brownish stones (**A**, ×400). Compared with uric acids stones (**B**, ×400), which typically have smooth surface and no prominent cracks (arrow heads), the stones had rough surface and cracks (typically seen in the stones indicated by arrows) which would reflect spicules radiating from the centre.

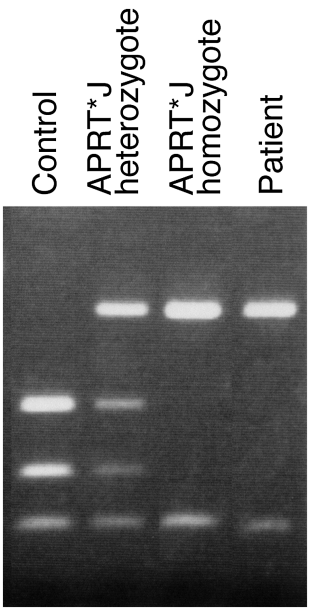


Fig. 2. PCR-based genomic analysis. After a restriction enzyme cut, the PCR product of the patient showed the same pattern as that of APRT*J homozygote in electrophoresis analysis.

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abolic pathway for adenine. The 2,8-DHA excreted into the urine is highly insoluble at any physiological pH, and its parenchymal and intratubular deposition may cause renal failure [2]. APRT deficiency may be an underdiagnosed cause of renal failure for which treatment is effective if renal damage has not progressed too far at diagnosis [1]. Skilful urinalysis to recognize the 2,8-DHA stones would be the most important diagnostic procedure.

Conflict of interest statement. None declared.

References

1. Edvardsson V, Palsson R, Olafsson I *et al*. Clinical features and genotype of adenine phosphoribosyltransferase deficiency in Iceland. *Am J Kidney Dis* 2001; 38: 473–480
2. Gelb AB, Fye KH, Tischfield JA *et al*. Renal insufficiency secondary to 2,8-dihydroxyadenine urolithiasis. *Hum Pathol* 1992; 23: 1081–1085
3. Kamatani N, Hakoda M, Otsuka S *et al*. Only three mutations account for almost all defective alleles causing adenine phosphoribosyltransferase deficiency in Japanese patients. *J Clin Invest* 1992; 90: 130–135

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